Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/05/2005
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Project Title:	Virtual Reality Training: Cybersickness and Effects of Sensorimotor Functions		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Task Description:	Though often brilliantly engineered, two critical and unresolved human factors issues in VRs are: 1) potential "cybersickness", a form of space motion sickness, which is experienced in virtual worlds, and 2) maladaptive sensorimotor responses following virtual environment (VE) training. Although, numerous studies show motion sickness and disturbances in sensorimotor functions following flight simulator training, during and following space flight, similar research concerning the aftereffects of VR training is only beginning to appear in literature. The plasticity of the human central nervous system makes it likely that users will eventually adapt to any unpleasant perceptual experiences in a virtual environment. However, the most critical problem for training applications is that behaviors learned in the VE may not be similar to the responses required in the real environment, potentially resulting the acquisition of inappropriate perceptual-motor behaviors. The purpose of this study is to evaluate and compare responses to two types of VR delivery systems (head-mounted display [HMD] and a dome-projection system), two exposure duration periods (30 minand 60 min), and repeated exposures (three sessions). Specific responses that we will examine include cybersickness severity and symptom patterns, task performance, and several sensorimotor functions (eye-head and eye-hand coordination, postural equilibrium). The overall goal of the proposed research is to examine the extent to which "immersion" in different types of VR systems produces motion sickness and disrupts perceptual and sensorimotor functions that may result in negative impacts on performance of operational activities. A secondary goal, although equally important, is to determine the extent to which VR aftereffects are similar sensorimotor adaptive responses to space flight. This information will provide support for the use of virtual environments as ground-based analogues for neurovestibular, sensorimotor and perceptual responses to space flight, and for the development of countermeasures using VR, both of which are high priority issues described in the Bioastronautics Critical Path Roadmap.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall goal of the proposed research is to examine the extent to which "immersion" in VR systems produces motion sickness and disrupts perceptual and sensorimotor functions that may ultimately result in negative impacts on performance of operational activities. Information derived from this work will be useful to a broad range of users and VR developers in entertainment, education, industry, and military in determining recommendations for safe usage of VR, such as implementation of precautionary measures for personal and work acitivities following immersion in VR.
Task Progress:	We completed data collection with 37/40 subjects (20 men and 17 women) performing a sensorimotor task in either the HMD or dome virtual reality system. This study is designed to examine the effect of two types of VR systems(HMD and dome), duration of exposure to VR (30min or 60min) and repeated exposures (3 separate days) to virtual reality on postural equilibrium. Postural equilibrium was measured using the Neurocom Equitest system sensory organization tests (SOTs). Preliminary analyses showed a significant decrease in overall postural control following exposure to virtual reality which returned to pre-exposure values over a 2-4 hour period, with the largest recovery in the first hour. Exposure to the dome system produced significantly larger decrements in postural stability than the HMD. The data also showed that postural control following virtual reality significantly improved across the three days, indicating subjects adapted over repeated exposures to virtual reality. Finally, the data showed that women did not exhibit significant decrements in posture following VR, whereas men did show significant decrements in postural equilibrium scores following VR exposure. Motion sickness improved significantly across the three days of repeated exposures and generally recovered within the first hour following VR exposure. We will be completing data collection for this study within about 1 month. The next study will begin immediately and will follow the same protocol with one exception. Measures of eye-head and eye-hand coordination will be evaluated before and after exposure to VR.
Bibliography Type:	Description: (Last Updated: 11/17/2010)
Articles in Peer-reviewed Journals	Stroud KJ, Harm DL, Klaus DM. "Preflight virtual reality training as a countermeasure for space motion sickness and disorientation." Aviat Space Environ Med. 2005 Apr;76(4):352-6. PMID: 15828634 , Apr-2005
Dissertations and Theses	Stroud KJ. "Mitigating vestibular disturbances during space flight using virtual reality training and reentry vehicle design guidelines." University of Colorado at Boulder, December 2004. , Dec-2004
Papers from Meeting Proceedings	Harm DL, Taylor LC. "Effects of exposure to HMD and dome VR systems on postural control." 11th International Conference on Human-Computer Interaction, Las Vegas, NV, July 2005. 11th International Conference on Human-Computer Interaction, Las Vegas, NV, July 2005. , Jul-2005
Presentation	Harm, D.L.; Taylor, L.C. "). Effects of exposure to virtual reality on postural control: Potential analog for space flight" 25th Annual International Gravitational Physiology Meeting Jun-2004
Presentation	Harm, D.L.; Taylor, L.C. "). Postural control disturbances produced by exposure to HMD and dome VR systems" Bioastronautics Investigators' Workshop Jan-2005